

Announcements

- Assignment coming up

Today

- More continuous time
- Exponential distribution
- Poisson process
- Poisson distribution

When does the individual die?

- Discrete approximation to continuous time
 - small interval of time, Δt
 - constant probability of death, p_{death}
- Set time interval very small
 - $\Delta t \rightarrow 0$

When does the individual die?

Algorithm

set $m, \Delta t, p_{\text{death}} = m\Delta t$

set state = alive, $t = 0$

while state = alive

$t = t + \Delta t$

 if death occurs given Bernoulli(p_{death})

 state = dead

print t (time to death)

Bernoulli process

Poisson process

- A continuous time process
- Waiting time between events is exponential
- Event sequence diagram

Poisson distribution

- Emerges as the sum of a Poisson process
- Number of events that occur in t